

# Reference Material: Numerical Analysis

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## INFORMATION

This is an informal note.

### About me

1. Hayata Yamasaki
2. Our research explores the fundamental theory of quantum information and quantum computation, along with their applications. Our goal is to establish a comprehensive theoretical foundation for the era of quantum

information processing enabled by quantum computers. For more details about our research, please visit <https://www.hayatayamasaki.com/>

3. If you have any questions, feel free to email me. Office hours are by appointment.

### Textbooks

1. The goal of this course is to acquire the material covered in the textbook [1].
2. Additional advanced topics can be found in another textbook [2].
3. For supplementary references, see also Refs. [3, 4].

## I. GAUSSIAN ELIMINATION AND LU DECOMPOSITION

### A. A system of linear equations

A system of  $N$  linear equations in  $N$  variables  $x_1, \dots, x_N$  is a set of equations in the form of

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,N}x_N &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,N}x_N &= b_2, \\ &\vdots \\ a_{N,1}x_1 + a_{N,2}x_2 + \dots + a_{N,N}x_N &= b_N, \end{aligned} \tag{1}$$

where  $a_{1,1}, \dots, a_{N,N}$  and  $b_1, \dots, b_N$  are numbers (for example, in  $\mathbb{R}$  or  $\mathbb{C}$ ). A solution of (1) is a tuple  $x_1, \dots, x_N$  of numbers that satisfy all the  $N$  equations. Equivalently, the system (1) of linear equations is written as

$$Ax = b, \tag{2}$$

where

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,N-1} & a_{1,N} \\ a_{2,1} & a_{2,2} & \dots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \dots & a_{N-1,N-1} & a_{N-1,N} \\ a_{N,1} & a_{N,2} & \dots & a_{N,N-1} & a_{N,N} \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{N-1} \\ x_N \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_{N-1} \\ b_N \end{pmatrix}. \tag{3}$$

Given an  $N \times N$  matrix  $A$ , the inverse of the matrix  $A$  is an  $N \times N$  matrix  $A^{-1}$  satisfying

$$A^{-1}A = AA^{-1} = I, \tag{4}$$

where  $I$  is the identity matrix. The inverse  $A^{-1}$  does not always exist, and its existence will be characterized through the rank and the determinant. The rank of a matrix  $A$  is the number of linearly independent row vectors of  $A$ , denoted by

$$\text{rank } A. \tag{5}$$

To define the determinant of an  $N \times N$  matrix  $A$ , we let  $A_{j,k}$  denote an  $(N-1) \times (N-1)$  submatrix obtained by deleting the  $j$ th row and the  $k$ th column of  $A$ . We define the determinant of  $A$  by induction. For  $N = 1$ , the determinant of  $A = (a_{1,1})$  is defined as

$$\det(A) := a_{1,1}. \tag{6}$$

For  $N \geq 2$ , the determinant of  $A$  is defined as

$$\det(A) := a_{j,1}c_{j,1} + a_{j,2}c_{j,2} + \dots + a_{j,N}c_{j,N}, \tag{7}$$

where we write

$$c_{j,k} := (-1)^{j+k} m_{j,k}, \quad (8)$$

$$m_{j,k} := \det(A_{j,k}). \quad (9)$$

Here, the determinant  $M_{j,k}$  of the submatrix  $A_{j,k}$  is called a minor, and  $C_{j,k}$  obtained by multiplying the minor by  $(-1)^{j+k}$  is called a cofactor.

An  $N \times N$  matrix  $A$  has its inverse  $A^{-1}$  if and only if

$$\text{rank } A = N, \quad (10)$$

or equivalently,

$$\det(A) \neq 0. \quad (11)$$

A matrix  $A$  that has its inverse  $A^{-1}$  is called an invertible matrix, or a non-singular matrix; otherwise, it is called a non-invertible matrix, or a singular matrix. In this case, using a cofactor matrix

$$C := \begin{pmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,N-1} & c_{1,N} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,N-1} & c_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{N-1,1} & c_{N-1,2} & \cdots & c_{N-1,N-1} & c_{N-1,N} \\ c_{N,1} & c_{N,2} & \cdots & c_{N,N-1} & c_{N,N} \end{pmatrix}, \quad (12)$$

the inverse is given by

$$A^{-1} := \frac{1}{\det(A)} C^\top, \quad (13)$$

where  $C^\top$  is the transpose of  $C$ . With  $A^{-1}$ , the solution of (2) is given by

$$x = A^{-1}b. \quad (14)$$

On the other hand, by induction from (7), one can show that the determinant  $\det(A)$  of an  $N \times N$  matrix is a degree- $N$  polynomial of  $N!$  terms. The Stirling's approximation yields

$$\sqrt{2\pi N} \left(\frac{N}{e}\right)^N e^{\frac{1}{12N+1}} \leq N! \leq \sqrt{2\pi N} \left(\frac{N}{e}\right)^N e^{\frac{1}{12N}} = \exp[\Theta(N \log N)], \quad (15)$$

making the computation of  $\det(A)$  from the definition infeasible, so we need a better technique.

## B. Gaussian elimination for solving a system of linear equations

The key technique for solving a system of  $N$  linear equations within polynomial time in  $N$  is to use triangular matrices. An  $N \times N$  matrix  $U$  (and  $L$ ) is called an upper triangular matrix (and a lower triangular matrix, respectively) if

$$U = \begin{pmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,N-1} & u_{1,N} \\ 0 & u_{2,2} & \cdots & u_{2,N-1} & u_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & u_{N-1,N-1} & u_{N-1,N} \\ 0 & 0 & \cdots & 0 & u_{N,N} \end{pmatrix}, \quad L = \begin{pmatrix} l_{1,1} & 0 & \cdots & 0 & 0 \\ l_{2,1} & l_{2,2} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ l_{N-1,1} & l_{N-1,2} & \cdots & l_{N-1,N-1} & 0 \\ l_{N,1} & l_{N,2} & \cdots & l_{N,N-1} & l_{N,N} \end{pmatrix}. \quad (16)$$

Gaussian elimination is a method to rewrite (2) into an equivalent form using a triangular matrix within polynomial time in  $N$ . Once Gaussian elimination is performed, the solution can be easily obtained by back substitution.

Example [1]: Solve a system of linear equations

$$\begin{pmatrix} 3 & 5 & 2 \\ 6 & 2 & 8 \\ 0 & 8 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ 26 \\ -7 \end{pmatrix}. \quad (17)$$

Gaussian elimination: For each  $n \in \{1, 2, \dots, N-1\}$ , choose a pivot, i.e., an element  $a_{n,k}$  that maximizes its absolute value  $|a_{n,k}|$  ( $k \geq n$ ), permute the  $n$ th row and the  $k$ th row, and eliminate remaining  $a_{n,k}$  for all  $k > n$  by adding and subtracting rows with appropriate scalar multiplied. Choosing the pivot of the largest absolute value is important for numerical stability.

1. For the 1st row:

(a) Pivoting:

$$\begin{pmatrix} 6 & 2 & 8 \\ 3 & 5 & 2 \\ 0 & 8 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ 8 \\ -7 \end{pmatrix}; \quad (18)$$

(b) Elimination:

$$\begin{pmatrix} 6 & 2 & 8 \\ 0 & 4 & -2 \\ 0 & 8 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ -5 \\ -7 \end{pmatrix}; \quad (19)$$

2. For the 2nd row:

(a) Pivoting:

$$\begin{pmatrix} 6 & 2 & 8 \\ 0 & 8 & 2 \\ 0 & 4 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ -7 \\ -5 \end{pmatrix}; \quad (20)$$

(b) Elimination:

$$\begin{pmatrix} 6 & 2 & 8 \\ 0 & 8 & 2 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ -7 \\ -3/2 \end{pmatrix}; \quad (21)$$

Back substitution: For each  $n \in \{N, N-1, \dots, 1\}$ , obtain  $x_N$  by substituting  $x_n, \dots, x_N$  in the  $n$ th row.

1. For the 3rd row:

$$\begin{pmatrix} 6 & 2 & 8 \\ 0 & 8 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ -7 \\ 1/2 \end{pmatrix}; \quad (22)$$

2. For the 2nd row:

$$\begin{pmatrix} 6 & 2 & 8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 26 \\ -1 \\ 1/2 \end{pmatrix}; \quad (23)$$

3. For the 1st row:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 1/2 \end{pmatrix}. \quad (24)$$

The runtime of Gaussian elimination is bounded as follows.

**Theorem 1** (Operation count of Gaussian elimination). *For any  $N \times N$  matrix  $A$  and any column vector  $b$  of  $N$  elements, Gaussian elimination uses  $O(N^3)$  arithmetic operations, as  $N \rightarrow \infty$ .*

*Proof.* For each  $n \in \{1, \dots, N\}$ , to process the  $n$ th row,

1. the pivoting step uses  $O(N)$  operations to permute rows;

2. the elimination for  $(N - n)$  rows with at most  $(N - n + 1)$  nonzero elements uses  $O((N - n)(N - n + 1))$  operations.

In total, the number of arithmetic operations is bounded by

$$\sum_{n=1}^{N-1} (O(N) + O((N - n)(N - n + 1))) = O(N^3). \quad (25)$$

□

### C. Gauss-Jordan elimination for matrix inversion

Given an  $N \times N$  matrix  $A$ , the task of matrix inversion is to compute each element of an  $N \times N$  matrix  $X$  satisfying  $AX = XA = I$ , where  $I$  is the identity matrix. The solution  $X$  is the inverse  $A^{-1}$  of  $A$  by definition. This task differs from solving a system of linear equations.

We can solve the matrix inversion by solving multiple systems of linear equations. In particular, let  $x_{*,1}, x_{*,2}, \dots, x_{*,N}$  be column vectors of  $N$  elements representing the columns of  $X$ , i.e.,

$$X = (x_{*,1}, x_{*,2}, \dots, x_{*,N}), \quad (26)$$

where

$$x_{*,1} = \begin{pmatrix} x_{1,1} \\ x_{2,1} \\ \vdots \\ x_{N,1} \end{pmatrix}, x_{*,2} = \begin{pmatrix} x_{1,2} \\ x_{2,2} \\ \vdots \\ x_{N,2} \end{pmatrix}, \dots, x_{*,N} = \begin{pmatrix} x_{1,N} \\ x_{2,N} \\ \vdots \\ x_{N,N} \end{pmatrix}. \quad (27)$$

We write

$$b_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, b_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, b_N = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \quad (28)$$

Then,  $X$  can be obtained by solving

$$Ax_{*,1} = b_1, \quad (29)$$

$$Ax_{*,2} = b_2, \quad (30)$$

$$\vdots \quad (31)$$

$$Ax_{*,N} = b_N. \quad (32)$$

Gauss-Jordan elimination is the algorithm to solve these systems of linear equations simultaneously. In particular, by using the same procedure as the Gaussian elimination applied for each row of  $A$  and the corresponding row on the right-hand side, one can convert

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} & a_{N-1,N} \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N-1} & a_{N,N} \end{pmatrix} \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,N-1} & x_{1,N} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,N-1} & x_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{N-1,1} & x_{N-1,2} & \cdots & x_{N-1,N-1} & x_{N-1,N} \\ x_{N,1} & x_{N,2} & \cdots & x_{N,N-1} & x_{N,N} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \quad (33)$$

into an equivalent form of

$$\begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,N-1} & x_{1,N} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,N-1} & x_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{N-1,1} & x_{N-1,2} & \cdots & x_{N-1,N-1} & x_{N-1,N} \\ x_{N,1} & x_{N,2} & \cdots & x_{N,N-1} & x_{N,N} \end{pmatrix} = \begin{pmatrix} ? & ? & \cdots & ? & ? \\ ? & ? & \cdots & ? & ? \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ ? & ? & \cdots & ? & ? \\ ? & ? & \cdots & ? & ? \end{pmatrix} \quad (34)$$

where the matrix appearing on the right-hand side becomes the inverse  $A^{-1}$  of  $A$ .

For example, solve the matrix inversion of the following matrix:

$$\begin{pmatrix} 3 & 5 & 2 \\ 6 & 2 & 8 \\ 0 & 8 & 2 \end{pmatrix}. \quad (35)$$

Solution: The inverse is given by

$$\begin{pmatrix} 5/12 & -1/24 & -1/4 \\ 1/12 & -1/24 & 1/12 \\ -1/3 & 1/6 & 1/6 \end{pmatrix}. \quad (36)$$

#### D. LU decomposition

Given an  $N \times N$  matrix  $A$ , the LU decomposition of the matrix  $A$  is

$$A = LU, \quad (37)$$

where  $L$  is a lower triangular matrix, and  $U$  is an upper triangular matrix. By permuting the rows of  $A$  as in the pivoting in the Gaussian elimination, one can always find an LU decomposition of  $A$  up to the row permutation. When obtaining the LU decomposition, one can always assume either of the following:

1. all diagonal elements of  $L$  is 1, or
2. all diagonal elements of  $U$  is 1.

Then, LU decomposition can be obtained by comparing the elements of the matrix.

Correspondingly, there are two methods for LU decomposition.

1. The Doolittle's methods: compare the matrix elements in

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} & a_{N-1,N} \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N-1} & a_{N,N} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ l_{2,1} & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ l_{N-1,1} & l_{N-1,2} & \cdots & 1 & 0 \\ l_{N,1} & l_{N,2} & \cdots & l_{N,N-1} & 1 \end{pmatrix} \begin{pmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,N-1} & u_{1,N} \\ 0 & u_{2,2} & \cdots & u_{2,N-1} & u_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & u_{N-1,N-1} & u_{N-1,N} \\ 0 & 0 & \cdots & 0 & u_{N,N} \end{pmatrix}. \quad (38)$$

2. The Crout's methods: compare the matrix elements in

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} & a_{N-1,N} \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N-1} & a_{N,N} \end{pmatrix} = \begin{pmatrix} l_{1,1} & 0 & \cdots & 0 & 0 \\ l_{2,1} & l_{2,2} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ l_{N-1,1} & l_{N-1,2} & \cdots & l_{N-1,N-1} & 0 \\ l_{N,1} & l_{N,2} & \cdots & l_{N,N-1} & l_{N,N} \end{pmatrix} \begin{pmatrix} 1 & u_{1,2} & \cdots & u_{1,N-1} & u_{1,N} \\ 0 & 1 & \cdots & u_{2,N-1} & u_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & u_{N-1,N} \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}. \quad (39)$$

For a positive semidefinite matrix, one can obtain a variant of LU decomposition called the Cholesky decomposition. The matrix  $A$  is called a positive definite matrix if for all column vectors  $v \neq 0$ , it holds that

$$v^\dagger A v > 0, \quad (40)$$

while  $A$  is called a positive semidefinite matrix if  $v^\dagger A v \geq 0$  for all  $v$ . Given a positive semidefinite matrix  $A$ , the Cholesky decomposition is

$$A = LL^\dagger, \quad (41)$$

where  $L$  is a lower triangular matrix, and  $L^\dagger$  is the conjugate transpose of  $L$  (which is equivalent to the tranpose of  $L$  when  $L$  is a real matrix, while may be different when  $L$  is a complex matrix). The main diagonals of  $L$  in the Cholesky matrix can be taken as real numbers. To obtain the Cholesky decomposition of  $A$ , solve

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} & a_{N-1,N} \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N-1} & a_{N,N} \end{pmatrix} = \begin{pmatrix} l_{1,1} & 0 & \cdots & 0 & 0 \\ l_{2,1} & l_{2,2} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ l_{N-1,1} & l_{N-1,2} & \cdots & l_{N-1,N-1} & 0 \\ l_{N,1} & l_{N,2} & \cdots & l_{N,N-1} & l_{N,N} \end{pmatrix} \begin{pmatrix} l_{1,1} & \overline{l_{1,2}} & \cdots & \overline{l_{1,N-1}} & \overline{l_{1,N}} \\ 0 & l_{2,2} & \cdots & \overline{l_{2,N-1}} & \overline{l_{2,N}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & l_{N-1,N-1} & \overline{l_{N-1,N}} \\ 0 & 0 & \cdots & 0 & l_{N,N} \end{pmatrix}, \quad (42)$$

where the main diagonals on the right-hand side are real numbers.

LU decomposition has the following applications.

1. Once  $A = LU$  is computed, the equation  $Ax = b$  can be solved by considering

$$Ly = b, \quad (43)$$

$$Ux = y, \quad (44)$$

which is straightforward to solve by back substitution since  $L$  and  $U$  are triangular. This is particularly useful when one needs to solve multiple different systems of linear equations for the same  $A$  and different  $b$ .

2. Once  $A = LU$  is computed, the determinant  $\det(A)$  can be computed by

$$\det(A) = \det(L) \det(U), \quad (45)$$

where the determinants of triangular matrices are the product of their diagonal elements, even if there are nonzero off-diagonal elements.

3. Once  $A = LU$  is computed, the inverse  $A^{-1}$  can be computed by

$$A^{-1} = U^{-1}L^{-1}, \quad (46)$$

where the Gauss-Jordan elimination for triangular matrices is straightforward by back substitution.

For example, obtain the LU decomposition of the following matrix:

$$\begin{pmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{pmatrix}. \quad (47)$$

Solution: The LU decomposition is obtained from

$$\begin{pmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{2,1} & 1 & 0 \\ l_{3,1} & l_{3,2} & 1 \end{pmatrix} \begin{pmatrix} u_{1,1} & u_{2,1} & u_{3,1} \\ 0 & u_{2,2} & u_{3,2} \\ 0 & 0 & u_{3,3} \end{pmatrix} = \begin{pmatrix} u_{1,1} & u_{2,1} & u_{3,1} \\ l_{2,1}u_{1,1} & l_{2,1}u_{2,1} + u_{2,2} & l_{2,1}u_{3,1} + u_{3,2} \\ l_{3,1}u_{1,1} & l_{3,1}u_{2,1} + l_{3,2}u_{2,2} & l_{3,1}u_{3,1} + l_{3,2}u_{3,2} + u_{3,3} \end{pmatrix}. \quad (48)$$

By comparing each matrix element on both sides, one obtains

$$\begin{pmatrix} u_{1,1} & u_{2,1} & u_{3,1} \\ 0 & u_{2,2} & u_{3,2} \\ 0 & 0 & u_{3,3} \end{pmatrix} = \begin{pmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 0 & 0 & 6 \end{pmatrix}, \quad (49)$$

$$\begin{pmatrix} 1 & 0 & 0 \\ l_{2,1} & 1 & 0 \\ l_{3,1} & l_{3,2} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}. \quad (50)$$

## II. CONDITION NUMBER AND STABILITY

### A. Well-conditioning and ill-conditioning

We have learned how to numerically solve

$$Ax = b, \quad (51)$$

where  $A$  is an  $N \times N$  matrix,  $b$  is a column vector of  $N$  elements, and  $x$  is a column vector representing unknown variables. In numerics, each number is represented up to a finite precision, so it is important to evaluate how the inaccuracy in inputs  $A$  and  $b$  to the algorithm affects the errors in the output solution  $x$ .

A computational problem is called ill-conditioned (or ill-posed) if small changes in the input cause large changes in the output. On the other hand, the problem is called well-conditioned (or well-posed) if small changes cause only small changes in the output. For example, consider

$$\begin{pmatrix} 0.9999 & -1.0001 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (52)$$

which yields a solution

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix}. \quad (53)$$

With a small error  $\epsilon > 0$  in the input, one may have

$$\begin{pmatrix} 0.9999 & -1.0001 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 + \epsilon \end{pmatrix}, \quad (54)$$

which yields a solution

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0.5 + 5000.5\epsilon \\ -0.5 + 4999.5\epsilon \end{pmatrix}. \quad (55)$$

In this case, a small input change of magnitude  $\epsilon$  leads to approximately  $5000\epsilon$  changes in the output.

We want to analyze such numerical stability in more general. For a system of linear equations  $Ax = b$ , consider a case where  $b$  is changed into  $\tilde{b}$ . In this case, one may obtain an approximate solution  $\tilde{x}$  of

$$A\tilde{x} = \tilde{b}. \quad (56)$$

The residual is defined as

$$r := b - \tilde{b} \quad (57)$$

$$= b - A\tilde{x} \quad (58)$$

$$= A(x - \tilde{x}), \quad (59)$$

representing the change in  $b$ . To obtain a solution  $\tilde{x}$  with high accuracy,  $r$  is necessarily small under reasonable assumptions, but even if  $r$  is small, the change in the output solutions may be much larger, as we have seen above. To analyze this effect quantitatively, we will use the tools of matrix norms and condition numbers.

### B. Matrix norm

Norms and distances are useful concepts when one wants to argue the magnitude of errors in numerics. Let  $\mathcal{V}$  be a vector space, which is closed under

1. addition  $u + v \in \mathcal{V}$  for any  $u, v \in \mathcal{V}$ , and
2. scalar multiplication  $\alpha v \in \mathcal{V}$  for any scalar  $\alpha$  (typically in  $\mathbb{R}$  or  $\mathbb{C}$ ) and  $v \in \mathcal{V}$ ;

A norm  $\|\cdot\|$  is a function from a vector space to the set of non-negative real numbers satisfying the following properties:

1. positive definiteness:  $\|v\| \geq 0$  and

$$\|v\| = 0 \Leftrightarrow v = 0; \quad (60)$$

2. subadditivity (triangle inequality):

$$\|u + v\| \leq \|u\| + \|v\|; \quad (61)$$

3. absolute homogeneity:

$$\|\alpha v\| = |\alpha| \|v\|. \quad (62)$$

This represents the magnitude of vectors.

For example, the  $l_1$ -norm is

$$\|v\|_1 := \sum_{n=1}^N |v_n|, \quad (63)$$

where

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{pmatrix}. \quad (64)$$

The  $l_2$ -norm (also known as the Euclidean norm) is

$$\|v\|_2 := \sqrt{\sum_{n=1}^N v_n^2}. \quad (65)$$

The  $l_\infty$ -norm is

$$\|v\|_\infty := \max_{n \in \{1, \dots, N\}} |v_n|. \quad (66)$$

In general, for any  $p \geq 1$ , the  $p$ -norm is defined as

$$\|v\|_p := \left( \sum_{n=1}^N |v_n|^p \right)^{1/p}. \quad (67)$$

A distance  $d$  is a function satisfying

1. positive definiteness:  $d(u, v) \geq 0$ , and  $d(u, v) = 0 \Leftrightarrow u = v$ ;
2. triangle inequality:  $d(u, v) \leq d(u, w) + d(w, v)$ ;
3. symmetry:  $d(u, v) = d(v, u)$ .

For vectors  $v$  and  $\tilde{v}$ , the norm induces a distance as

$$d(v, \tilde{v}) = \|v - \tilde{v}\|. \quad (68)$$

The set of  $N \times N$  matrices is also a vector space; indeed, it is closed under addition and scalar multiplication. A canonical choice of a norm of an  $N \times N$  matrix  $A$  is a matrix norm given by

$$\|A\| := \sup \{\|Av\| : \|v\| \leq 1\}, \quad (69)$$

where  $\|v\|$  is a norm of the vector space for  $v$ . Note that  $\|A\|$  depends on the vector norm  $\|v\|$  selected.

Matrix norms are closely related to the eigenvalues. For  $N \times N$  complex matrices, we have the following definitions.

1. An  $N \times N$  complex matrix  $N$  is called a normal matrix if

$$N^\dagger N = N N^\dagger, \quad (70)$$

where  $N^\dagger$  is the conjugate transpose of  $N$ .

2. An  $N \times N$  complex matrix  $H$  is called a hermitian matrix if

$$H^\dagger = H. \quad (71)$$

3. An  $N \times N$  complex matrix  $U$  is called a unitary matrix if

$$U^\dagger U = U U^\dagger = I, \quad (72)$$

where  $I$  is the identity matrix. In particular, an  $N \times N$  real matrix  $O$  is called an orthogonal matrix if

$$O^\top O = O O^\top = I, \quad (73)$$

which is a special case of complex unitary matrices.

4. An  $N \times N$  complex matrix  $A$  is called a positive semidefinite matrix if for all vector  $v$

$$v^\dagger A v \geq 0. \quad (74)$$

5. An  $N \times N$  complex matrix  $A$  is called a positive definite matrix if for all vector  $v$

$$v^\dagger A v > 0. \quad (75)$$

6. An  $N \times N$  complex matrix  $\rho$  is called a density matrix if  $\rho$  is positive semidefinite and

$$\text{Tr}[\rho] := \sum_{n=1}^N e_n^\dagger \rho e_n = 1, \quad (76)$$

where  $\{e_n\}_{n=1,\dots,N}$  is the standard basis.

For a matrix  $A$ , suppose that one has a scalar  $\lambda$  and a nonzero vector  $v \neq 0$  satisfying

$$A v = \lambda v; \quad (77)$$

then,  $\lambda$  is called an eigenvalue of  $A$ , and  $v$  is called an eigenvector of  $A$  corresponding to the eigenvalue  $\lambda$ . For an  $N \times N$  complex matrix  $A$ , its spectral decomposition

$$A = \sum_{j=1}^J \lambda_j \Pi_j, \quad (78)$$

where  $\lambda_j \in \mathbb{C}$  is an eigenvalue, and  $\Pi_j$  is the projection matrix onto the set of eigenvectors corresponding to  $\lambda_j$ . The matrix  $A$  has a spectral decomposition if and only if  $A$  is a normal matrix. For a function  $f : \mathbb{C} \rightarrow \mathbb{C}$  and a matrix  $A$  with the spectral decomposition (78), we define an operator function as

$$f(A) := \sum_{j=1}^J f(\lambda_j) \Pi_j. \quad (79)$$

For example, we have

$$|A| = \sum_{j=1}^J |\lambda_j| \Pi_j, \quad (80)$$

$$\sqrt{A} = \sum_{j=1}^J \sqrt{\lambda_j} \Pi_j. \quad (81)$$

The examples of the matrix norms include the following.

1. Using the  $l_2$ -norm  $\|v\|_2$ , we have the spectral norm  $\|A\| = \sqrt{\lambda_{\max}(A^\dagger A)}$ , where  $A^\dagger$  is the conjugate transpose matrix of  $A$  (which equals the transpose matrix for a real-valued matrix, but the complex conjugate is further taken for a complex-valued matrix), and  $\lambda_{\max}$  represents the maximum eigenvalue;
2. Using the  $l_1$ -norm  $\|v\|_1$ , we have the column sum norm  $\|A\| = \max_k \sum_{j=1}^N |a_{j,k}|$ ;
3. Using the  $l_\infty$ -norm  $\|v\|_\infty$ , we have the row sum norm  $\|A\| = \max_j \sum_{k=1}^N |a_{j,k}|$ .

Using a matrix norm  $\|A\|$ , one can evaluate the norm of a vector  $Ax$  as

$$\|Av\| \leq \|A\| \|v\|, \quad (82)$$

where  $\|v\|$  is the vector norm used for defining the matrix norm  $\|A\|$ . All these norms are submultiplicative

$$\|AA'\| \leq \|A\| \|A'\|. \quad (83)$$

The matrix norm is not the only way of defining the norm; for example, the Schatten  $p$ -norm is defined as

$$\|A\| = (\text{Tr}[|A|^p])^{1/p}. \quad (84)$$

The examples of the Schatten  $p$ -norms include the following.

1. If  $p = \infty$ , the  $p$ -norm coincides with the spectral norm  $\|A\| = \sqrt{\lambda_{\max}(A^\dagger A)}$ .
2. If  $p = 1$ , the  $p$ -norm yields the trace norm (also called the nuclear norm)  $\|A\|_{\text{tr}} = \text{Tr}[\sqrt{A^\dagger A}]$ .
3. if  $p = 2$ , the  $p$ -norm yields the Frobenius norm given by

$$\|A\|_F := \sum_{j=1}^N \sum_{k=1}^N |a_{j,k}|^2. \quad (85)$$

### C. Condition number

Consider a system of linear equations represented by

$$Ax = b, \quad (86)$$

where  $A$  is an  $N \times N$  non-singular matrix,  $b$  is a column vector of  $N$  elements, and  $x$  is a column vector representing unknown variables. Using a matrix norm  $\|A\|$ , the condition number is defined as

$$\kappa(A) := \|A\| \|A^{-1}\|. \quad (87)$$

When the matrix norm is the spectral norm, i.e., the vector norm used is the  $l_2$  norm, the condition number  $\kappa(A)$  is the ratio between the maximum absolute value and the minimum absolute value of eigenvalues of  $A$ .

The condition number is key to analyzing the ill-conditioning of the system of linear equations, as shown in the following theorem. The theorem means that the magnitude  $\|x - \tilde{x}\|$  of the changes in solution relative to  $\|x\|$  depends on the condition number  $\kappa(A)$  as well as the norm of the residual  $\|r\|$  relative to  $\|b\|$ .

**Theorem 2** (Well-conditioned linear systems with inaccurate vectors). *Supposed that one is given a system of linear equations*

$$Ax = b, \quad (88)$$

where  $A$  is an  $N \times N$  non-singular matrix,  $b \neq 0$  is a column vector of  $N$  elements, and  $x$  is a column vector representing unknown variables. Let  $\tilde{b}$  be a column vector approximating  $b$ , and  $\tilde{x}$  be a solution of

$$A\tilde{x} = \tilde{b}. \quad (89)$$

Then, it holds that

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leq \kappa(A) \frac{\|b - \tilde{b}\|}{\|b\|}, \quad (90)$$

where the denominator on the right-hand side represents the residual as in (57).

*Proof.* Due to  $b \neq 0$  and  $A$  is invertible, it holds that

$$x = A^{-1}b \neq 0. \quad (91)$$

Thus, due to (60), we have

$$\|b\| > 0, \|x\| > 0. \quad (92)$$

Due to (88) and (89), we have

$$A^{-1}(b - \tilde{b}) = A^{-1}(Ax - A\tilde{x}) \quad (93)$$

$$= A^{-1}A(x - \tilde{x}) \quad (94)$$

$$= x - \tilde{x}. \quad (95)$$

Thus, using (82), we obtain

$$\|x - \tilde{x}\| = \|A^{-1}(b - \tilde{b})\| \quad (96)$$

$$\leq \|A^{-1}\| \|b - \tilde{b}\|. \quad (97)$$

From (88), using (82), we obtain

$$\|b\| = \|Ax\| \quad (98)$$

$$\leq \|A\| \|x\|, \quad (99)$$

and hence

$$\frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|}, \quad (100)$$

where we used (92).

Therefore, (100) and (96) yield

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leq \frac{\|A\| \|A^{-1}\| \|b - \tilde{b}\|}{\|b\|} \quad (101)$$

$$= \kappa(A) \frac{\|b - \tilde{b}\|}{\|b\|}, \quad (102)$$

where the last line follows from the definition (87) of the condition number.  $\square$

**Theorem 3** (Well-conditioned linear systems with inaccurate matrices). *Supposed that one is given a system of linear equations*

$$Ax = b, \quad (103)$$

where  $A$  is an  $N \times N$  non-singular matrix,  $b \neq 0$  is a column vector of  $N$  elements, and  $x$  is a column vector representing unknown variables. Let  $\tilde{A}$  be an invertible matrix approximating  $A$ , and  $\tilde{x}$  be a solution of

$$\tilde{A}\tilde{x} = b. \quad (104)$$

Then, it holds that

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leq \kappa(\tilde{A}) \frac{\|A - \tilde{A}\|}{\|\tilde{A}\|}. \quad (105)$$

*Proof.* It holds that

$$0 = Ax - \tilde{A}\tilde{x} = (A - \tilde{A})x + \tilde{A}(x - \tilde{x}). \quad (106)$$

Thus, we have

$$x - \tilde{x} = -\tilde{A}^{-1}(A - \tilde{A})x. \quad (107)$$

Therefore, using (83) and (82), we obtain

$$\|x - \tilde{x}\| \leq \|\tilde{A}^{-1}\| \|A - \tilde{A}\| \|x\|, \quad (108)$$

which yields the conclusion

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leq \kappa(\tilde{A}) \frac{\|A - \tilde{A}\|}{\|\tilde{A}\|}. \quad (109)$$

□

### III. METHODS FOR SOLVING LARGE-SCALE SYSTEMS OF LINEAR EQUATIONS

#### A. Conjugate gradient method

The conjugate gradient method solves a system of linear equations  $Ax = b$  when  $A$  is a positive definite matrix, which is particularly efficient when  $A$  is sparse, i.e., has few nonzero elements. An inner product  $\langle \cdots, \cdots \rangle$  is a scalar-valued function satisfying the following properties:

1.  $\langle u, v \rangle = \overline{\langle v, u \rangle}$ ;
2.  $\langle u, \alpha_1 v_1 + \alpha_2 v_2 \rangle = \alpha_1 \langle u, v_1 \rangle + \alpha_2 \langle u, v_2 \rangle$ ;
3.  $\langle v, v \rangle \geq 0$ , and  $\langle v, v \rangle = 0 \Leftrightarrow v = 0$ .

For a positive definite matrix  $A$ , we define

$$\langle u, v \rangle_A := u^\dagger A v, \quad (110)$$

which is an inner product satisfying the above properties. The norm induced by the inner product  $\langle \cdots, \cdots \rangle_A$  is given by

$$\|v\|_A := \sqrt{\langle v, v \rangle_A} = \sqrt{v^\dagger A v} \quad (111)$$

In terms of this inner product, two vectors  $u, v$  are orthogonal if

$$\langle u, v \rangle_A = 0, \quad (112)$$

Also, in terms of the norm, a vector  $v$  is normalized if

$$\|v\|_A = 1. \quad (113)$$

For a vector space  $\mathcal{V}$ , a basis is a set  $\{v_1, v_2, \dots \in \mathcal{V}\}$  of vectors satisfying the following property:

1. the basis is linearly independent, i.e., for every finite subset  $\{v_1, v_2, \dots, v_M\}$ ,

$$\sum_{m=1}^M \alpha_m v_m = 0 \Leftrightarrow \alpha_1 = \alpha_2 = \cdots = \alpha_M = 0. \quad (114)$$

2. the basis is a spanning set, i.e., for any vector  $v \in \mathcal{V}$ , there exists  $\alpha_1, \alpha_2, \dots$  such that

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots. \quad (115)$$

The number of vectors in the basis is called the dimension of the vector space  $\mathcal{V}$ . In our case, we consider a finite-dimensional vector space  $\mathcal{V}$  with dimension  $N$ . If the vectors in a basis are orthogonal to each other, then the basis is called an orthogonal basis. If the vectors in a basis are orthogonal to each other and also normalized, then the basis is called an orthonormal basis. In general, the vectors in a basis are not necessarily orthogonal or normalized.

Given any basis  $\{v_1, \dots, v_N\}$ , the Gram-Schmidt orthogonalization is a way of constructing an orthogonal basis  $\{e_1, \dots, e_N\}$  from the given (non-orthogonal) basis. For any choice of inner product  $\langle \cdot, \cdot \rangle$ , the process proceeds as follows.

1. Set

$$e_1 := v_1; \quad (116)$$

2. For each  $n \in \{2, \dots, N\}$ , set

$$e_n := v_n - \sum_{j=1}^{n-1} \frac{\langle e_j, v_n \rangle}{\langle e_j, e_j \rangle} e_j. \quad (117)$$

The orghogonality of  $\{e_1, \dots, e_N\}$  is shown by induction: assuming  $\{e_1, \dots, e_{N-1}\}$  is orthogonal, one obtains, for all  $k \in \{1, \dots, N-1\}$ ,

$$\langle e_k, e_n \rangle = \left\langle e_k, v_n - \sum_{j=1}^{n-1} \frac{\langle e_j, v_n \rangle}{\langle e_j, e_j \rangle} e_j \right\rangle = 0 \quad (118)$$

The conjugate gradient method finds a solution  $x^*$  of  $Ax = b$  by iteratively constructing an orthogonal basis  $\{p_1, p_2, \dots, p_N\}$  in terms of the inner product  $\langle \dots, \dots \rangle_A$ , as in the Gram-Schmidt orthogonalization, and identifying the coefficients of the linear expansion of the solution  $x^*$  in the form of

$$x^* = \alpha_1 p_1 + \alpha_2 p_2 + \dots + \alpha_N p_N. \quad (119)$$

In particular, the conjugate gradient method starts from any initial point  $x^{(0)}$ , which is expanded as

$$x^{(0)} = \tilde{\alpha}_1 p_1 + \tilde{\alpha}_2 p_2 + \dots + \tilde{\alpha}_N p_N. \quad (120)$$

In the first step, it computes and uses the residual  $r_1$  to start the Gram-Schmidt orthogonalization with

$$r_1 = b - Ax^{(0)}, \quad (121)$$

$$p_1 = r_1. \quad (122)$$

The point is updated into

$$x^{(1)} = x^{(0)} + \frac{r_1^\dagger r_1}{p_1^\dagger A p_1} p_1. \quad (123)$$

For  $t \in \{2, 3, \dots, N\}$ , the algorithm in the  $t$ th iteration computes

$$r_{t+1} = r_t - \frac{r_t^\dagger r_t}{p_t^\dagger A p_t} A p_t, \quad (124)$$

$$p_{t+1} = r_{t+1} + \frac{r_{t+1}^\dagger r_{t+1}}{r_t^\dagger r_t} p_t,$$

$$x^{(t+1)} = x^{(t)} + \frac{r_{t+1}^\dagger r_{t+1}}{p_{t+1}^\dagger A p_{t+1}} p_{t+1},$$

outputting  $x^{(N)}$  in the end.

The conjugate gradient method has the following guarantee. In particular, it terminates in at most  $N$  iterations and outputs the solution  $x^{(N)} = x^*$ . As shown in the theorem, it computes the residual  $r_t$  for each iteration, and in practice, when the residual  $r_t$  is sufficiently small,  $x^{(t)}$  can be output as an approximate solution. In each iteration, the computational time is dominated by the computation of  $A p_t$ ,  $p_t^\dagger A p_t$ , and  $r_t^\dagger r_t$ , which is faster than the Gaussian elimination with  $O(N^3)$  arithmetic operations, most drastically when  $A$  is a sparse matrix.

**Theorem 4** (Conjugate gradient). *For each  $t \in \{1, 2, \dots, N\}$ , the conjugate gradient method in (124) achieves the following.*

1. (Residual) It holds that

$$r_t = b - Ax^{(t-1)}. \quad (125)$$

2. (Orthogonality of  $\{r_n\}_n$ ) It holds that

$$p_j^\dagger r_t = \begin{cases} 0 & \text{if } j \in \{1, \dots, t-1\}, \\ r_t^\dagger r_t & \text{if } j = t, \end{cases} \quad (126)$$

$$r_j^\dagger r_t = 0 \quad \text{for any } j \in \{1, \dots, t-1\}.$$

3. (Orthogonality of  $\{p_n\}_n$  in terms of  $\langle \cdots, \cdots \rangle_A$ ) For any  $j \in \{1, 2, \dots, t-1\}$ , it holds that

$$\begin{aligned} p_j^\dagger A r_t &= \begin{cases} 0 & \text{if } j \in \{1, \dots, t-2\}, \\ -\frac{r_t^\dagger r_t}{r_{t-1}^\dagger r_{t-1}} p_{t-1}^\dagger A p_{t-1} & \text{if } j = t-1, \end{cases} \\ p_j^\dagger A p_t &= 0. \end{aligned} \quad (127)$$

4. (Convergence) It holds that

$$x^{(t)} = \begin{cases} \sum_{n=1}^t \alpha_n p_n + \sum_{n=t+1}^N \tilde{\alpha}_n p_n & (t \in \{1, \dots, N-1\}), \\ \sum_{n=1}^N \alpha_n p_n = x^* & (t = N), \end{cases} \quad (128)$$

where  $\tilde{\alpha}_n$  and  $\alpha_n$  are coefficients of the initial point  $x^{(0)}$  and the solution  $x^*$  given, respectively, by (120) and (119).

*Proof.* The case of  $t = 1$ : It holds, by definition, that

$$r_1 = b - A x^{(0)}, \quad (129)$$

$$p_1^\dagger A r_1 = p_1^\dagger A p_1, \quad (130)$$

$$r_1^\dagger p_1 = r_1^\dagger r_1. \quad (131)$$

We also have

$$x^{(1)} = x^{(0)} + \frac{r_1^\dagger r_1}{p_1^\dagger A p_1} p_1 \quad (132)$$

$$= x^{(0)} + \frac{p_1^\dagger r_1}{p_1^\dagger A p_1} p_1 \quad (133)$$

$$= x^{(0)} + \frac{p_1^\dagger (b - A x^{(0)})}{p_1^\dagger A p_1} p_1 \quad (134)$$

$$= x^{(0)} + \frac{p_1^\dagger (A x^* - A x^{(0)})}{p_1^\dagger A p_1} p_1 \quad (135)$$

$$= x^{(0)} + (\alpha_1 - \tilde{\alpha}_1) p_1 \quad (136)$$

$$= \alpha_1 p_1 + \sum_{n=2}^N \tilde{\alpha}_n p_n. \quad (137)$$

The case of  $t \in \{2, \dots, N\}$ : Assume that the statement holds for  $t \in \{1, 2, \dots, T\}$ . We show the statement for  $t = T+1$  by induction.

Proof of (125): The residual satisfies

$$r_{T+1} = r_T - \frac{r_T^\dagger r_T}{p_T^\dagger A p_T} A p_T \quad (138)$$

$$= b - A x^{(T-1)} - \frac{r_T^\dagger r_T}{p_T^\dagger A p_T} A p_T \quad (139)$$

$$= b - A \left( x^{(T-1)} + \frac{r_T^\dagger r_T}{p_T^\dagger A p_T} p_T \right) \quad (140)$$

$$= b - A x^{(T)}. \quad (141)$$

Proof of (126): The first equation is obtained from the following case analysis.

1. For  $j \in \{1, \dots, T\}$ , we have

$$p_j^\dagger r_{T+1} = p_j^\dagger \left( r_T - \frac{r_T^\dagger r_T}{p_T^\dagger A p_T} A p_T \right) \quad (142)$$

$$= p_j^\dagger r_T - \frac{r_T^\dagger r_T}{p_T^\dagger A p_T} p_j^\dagger A p_T \quad (143)$$

$$= 0, \quad (144)$$

where we use the assumptions  $p_j^\dagger r_T = 0$  and  $p_j^\dagger A p_T = 0$  of induction.

2. For  $j = T + 1$ , we have

$$p_{T+1}^\dagger r_{T+1} = \left( r_{T+1}^\dagger + \frac{r_{T+1}^\dagger r_{T+1}}{r_T^\dagger r_T} p_T^\dagger \right) r_{T+1} \quad (145)$$

$$= r_{T+1}^\dagger r_{T+1} + \frac{r_{T+1}^\dagger r_{T+1}}{r_T^\dagger r_T} p_T^\dagger r_{T+1} \quad (146)$$

$$= r_{T+1}^\dagger r_{T+1}, \quad (147)$$

where we use  $p_T^\dagger r_{T+1} = 0$  shown above.

The second equation is via the following case analysis.

1. For  $j = 1$ , we have

$$r_1^\dagger r_{T+1} = p_1^\dagger r_{T+1} \quad (148)$$

$$= 0, \quad (149)$$

where we use  $p_1^\dagger r_{T+1} = 0$  shown above.

2. Otherwise, i.e., for  $j \in \{2, \dots, T\}$ , we have

$$r_j^\dagger r_{T+1} = \left( p_j^\dagger - \frac{r_j^\dagger r_j}{r_{j-1}^\dagger r_{j-1}} p_{j-1}^\dagger \right) r_{T+1} \quad (150)$$

$$= p_j^\dagger r_{T+1} - \frac{r_j^\dagger r_j}{r_{j-1}^\dagger r_{j-1}} p_{j-1}^\dagger r_{T+1} \quad (151)$$

$$= 0, \quad (152)$$

where we use  $p_j^\dagger r_{T+1}$  for  $j \in \{1, \dots, T\}$  shown above.

Proof of (127): The first equation is via the following case analysis

1. If  $j \in \{1, \dots, T - 1\}$ , we have

$$p_j^\dagger A r_{T+1} = -\frac{p_j^\dagger A p_j}{r_j^\dagger r_j} (r_{j+1}^\dagger - r_j^\dagger) r_{T+1} \quad (153)$$

$$= -\frac{p_j^\dagger A p_j}{r_j^\dagger r_j} (r_{j+1}^\dagger r_{T+1} - r_j^\dagger r_{T+1}) \quad (154)$$

$$= 0, \quad (155)$$

where we use  $r_j^\dagger r_{T+1} = 0$  for  $j \in \{1, \dots, T\}$  shown above.

2. For  $j = T$ , we have

$$p_T^\dagger A r_{T+1} = -\frac{p_T^\dagger A p_T}{r_T^\dagger r_T} (r_{T+1}^\dagger - r_T^\dagger) r_{T+1} \quad (156)$$

$$= -\frac{p_T^\dagger A p_T}{r_T^\dagger r_T} (r_{T+1}^\dagger r_{T+1} - r_T^\dagger r_{T+1}) \quad (157)$$

$$= -\frac{r_{T+1}^\dagger r_{T+1}}{r_T^\dagger r_T} p_T^\dagger A p_T, \quad (158)$$

where we use  $r_T^\dagger r_{T+1} = 0$  shown above.

The second equation follows from the Gram-Schmidt orthogonalization

$$p_{T+1} = r_{T+1} + \frac{r_{T+1}^\dagger r_{T+1}}{r_T^\dagger r_T} p_T \quad (159)$$

$$= r_{T+1} - \frac{p_T^\dagger A r_{T+1}}{p_T^\dagger A p_T} p_T \quad (160)$$

$$= r_{T+1} - \sum_{j=1}^T \frac{p_j^\dagger A r_{T+1}}{p_j^\dagger A p_j} p_j \quad (161)$$

$$= r_{T+1} - \sum_{j=1}^T \frac{\langle p_j, r_{T+1} \rangle_A}{\langle p_j, p_j \rangle_A} p_j, \quad (162)$$

where (160) follows from  $p_T^\dagger A r_{T+1} = -\frac{r_{T+1}^\dagger r_{T+1}}{r_T^\dagger r_T} p_T^\dagger A p_T$  shown above, and (161) from  $p_j^\dagger A r_{T+1} = 0$  for  $j \in \{1, \dots, T-1\}$  shown above. Since  $p_{T+1}$  is represented as the form of the Gram-Schmidt orthogonalization in (117), it holds, for all  $j \in \{1, \dots, T\}$ , that

$$p_j^\dagger A p_{T+1} = \langle p_j, p_{T+1} \rangle_A = 0. \quad (163)$$

Proof of (128): It holds that

$$x^{(T+1)} = x^{(T)} + \frac{r_{T+1}^\dagger r_{T+1}}{p_{T+1}^\dagger A p_{T+1}} p_{T+1} \quad (164)$$

$$= x^{(T)} + \frac{p_{T+1}^\dagger r_{T+1}}{p_{T+1}^\dagger A p_{T+1}} p_{T+1} \quad (165)$$

$$= x^{(T)} + \frac{p_{T+1}^\dagger (b - A x^{(T+1)})}{p_{T+1}^\dagger A p_{T+1}} p_{T+1} \quad (166)$$

$$= x^{(T)} + \frac{p_{T+1}^\dagger (A x^* - A x^{(T+1)})}{p_{T+1}^\dagger A p_{T+1}} p_{T+1} \quad (167)$$

$$= x^{(T)} + (\alpha_{T+1} - \tilde{\alpha}_{T+1}) p_{T+1} \quad (168)$$

$$= \begin{cases} \sum_{n=1}^{T+1} \alpha_n p_n + \sum_{n=T+2}^N \tilde{\alpha}_n p_n & (T \in \{1, \dots, N-2\}) \\ \sum_{n=1}^N \alpha_n p_n = x^* & (T = N-1), \end{cases} \quad (169)$$

where we use  $r_{T+1}^\dagger r_{T+1} = p_{T+1}^\dagger r_{T+1}$  shown above.  $\square$

## B. Gauss-Seidel method and Jacobi method

The methods for solving systems of  $N$  linear equations based on Gaussian elimination and LU decomposition are classified as direct methods, which use  $O(N^3)$  arithmetic operations to solve them by directly transforming the matrices. In contrast, an indirect (or iterative) method starts with an initial point and iteratively updates the points until the point converges to an approximation of the solution with a required accuracy. It is useful to apply indirect methods when each iteration is fact, and the convergence is rapid.

Consider a system of linear equations represented by

$$Ax = b, \quad (170)$$

where  $A$  is an  $N \times N$  invertible matrix. We decompose  $A$  into

$$A = D + L + U, \quad (171)$$

where  $D$  is a diagonal matrix, and  $L$  and  $U$  are, respectively, lower and upper triangular matrices with zero main diagonals. Below, iterative methods for solving this system of linear equations are introduced.

A canonical example of the iterative method is the Gauss-Seidel method. A solution  $x$  must satisfy

$$(D + L + U)x = b, \quad (172)$$

or equivalently,

$$x = (D + L)^{-1}(b - Ux). \quad (173)$$

The Gauss-Seidel method proceeds as follows:

1. an initial point  $x^{(0)}$  is chosen, such as  $x^{(0)} = 0$ ;
2. for each iteration  $t \in \{1, \dots, T\}$ , the point  $x^{(t-1)}$  is updated into

$$x^{(t)} = (D + L)^{-1}(b - Ux^{(t-1)}), \quad (174)$$

which can be computed, for each  $j \in \{1, \dots, N\}$ , by

$$x_j^{(t)} = \frac{1}{a_{j,j}} \left( b_j - \sum_{k=1}^{j-1} a_{j,k} x_k^{(t)} - \sum_{k=j+1}^N a_{j,k} x_k^{(t-1)} \right). \quad (175)$$

3. output  $x^{(T)}$  (in practice, check the convergence and terminate before the maximum number  $T$  of iterations).

To see that (174) and (175) are equivalent, consider solving  $(D + L)x^{(t)} = (b - Ux^{(t-1)})$  by back substitution of

$$\begin{pmatrix} a_{1,1} & 0 & \cdots & 0 & 0 \\ a_{2,1} & a_{2,2} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} & 0 \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N-1} & a_{N,N} \end{pmatrix} \begin{pmatrix} x_1^{(t)} \\ x_2^{(t)} \\ \vdots \\ x_{N-1}^{(t)} \\ x_N^{(t)} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_{N-1} \\ b_N \end{pmatrix} - \begin{pmatrix} 0 & a_{1,2} & \cdots & a_{1,N-1} & a_{1,N} \\ 0 & 0 & \cdots & a_{2,N-1} & a_{2,N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & a_{N-1,N} \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1^{(t-1)} \\ x_2^{(t-1)} \\ \vdots \\ x_{N-1}^{(t-1)} \\ x_N^{(t-1)} \end{pmatrix}. \quad (176)$$

A similar yet different example of the iterative method is the Jacobi method. The Jacobi method may converge more slowly than the Gauss-Seidel method, but it is more suitable for parallelization. Instead of (173), a solution  $x$  satisfies

$$x = D^{-1}(b - (L + U)x). \quad (177)$$

The Jacobi method proceeds as follows:

1. an initial point  $x^{(0)}$  is chosen, such as  $x^{(0)} = 0$ ;
2. for each iteration  $t \in \{1, \dots, T\}$ , the point  $x^{(t-1)}$  is updated into

$$x^{(t)} = D^{-1}(b - (L + U)x^{(t-1)}); \quad (178)$$

3. output  $x^{(T)}$  (in practice, check the convergence and terminate before the maximum number  $T$  of iterations).

Since (178) can be computed element-wise, i.e., independently for each  $x_j^{(t)}$  ( $j \in \{1, \dots, N\}$ ), this is more suitable for parallelization.

In the following, the condition for convergence and the required number of iterations in the Gauss-Seidel and Jacobi methods will be analyzed. Suppose that one is given a system of linear equations

$$Ax = b, \quad (179)$$

where  $A$  is an  $N \times N$  invertible matrix,  $b \neq 0$  is a column vector of  $N$  elements, and  $x$  is a column vector representing unknown variables. Let  $x^*$  denote the solution, which is assumed to satisfy, as in (173) and (177),

$$Mx^* = M'x^* + b, \quad (180)$$

where  $M$  is an invertible matrix. Fix an initial point  $x^{(0)}$ , and for each  $t \in \{1, \dots, T\}$ , consider an iteration given by

$$Mx^{(t)} = M'x^{(t-1)} + b. \quad (181)$$

Define an iteration matrix

$$C := M^{-1}M'. \quad (182)$$

In particular, in the Gauss-Seidel method, we have

$$C = (D + L)^{-1}U, \quad (183)$$

and, in the Jacobi method, we have

$$C = D^{-1}(L + U). \quad (184)$$

Using matrix norms, one can show the following error bound for this type of iterative method.

**Theorem 5** (Error bound in iterative methods). *For the iterative methods in (181), the point  $x^{(T)}$  after  $T$  iterations satisfies*

$$\|x^{(T)} - x^*\| \leq \|C\|^T \|x^{(0)} - x^*\|, \quad (185)$$

where  $C$  is the iterative matrix in (182), and  $\|C\|$  is the matrix norm of  $C$  for a vector norm for measuring accuracy  $\|x^{(t)} - x^*\|$ .

*Proof.* The convergence analysis is performed by constructing a recurrence formula on the error bound for each iteration. To this goal, the solution  $x^*$  satisfies. Hence, due to (181), we have

$$x^{(t)} - x^* = M^{-1}N(x^{(t-1)} - x^*) \quad (186)$$

$$= C(x^{(t-1)} - x^*). \quad (187)$$

By using this recurrence formula, we have

$$x^{(T)} - x^* = C(x^{(T-1)} - x^*) \quad (188)$$

$$= C^2(x^{(T-2)} - x^*) \quad (189)$$

$$\vdots \quad (190)$$

$$= C^T(x^{(0)} - x^*). \quad (191)$$

By taking the norm on both sides, we obtain

$$\|x^{(T)} - x^*\| = \|C^T(x^{(0)} - x^*)\| \leq \|C\|^T \|x^{(0)} - x^*\|, \quad (192)$$

where the last line used the submultiplicability (83) of the matrix norm.  $\square$

An immediate corollary of this theorem on the error bound is the following theoretical guarantee. In particular, the corollary shows that the Gauss-Seidel method converges if

$$\|(D + L)^{-1}U\| < 1, \quad (193)$$

and the Jacobi method converges if

$$\|D^{-1}(L + U)\| < 1. \quad (194)$$

Of particular importance are the cases where  $\|\dots\|$  is the spectral norm of matrices, i.e., where the left-hand sides of these conditions represent the maximum absolute values of the relevant matrices, called the spectral radius. These conditions hold if the matrix  $A$  is strictly diagonally dominant, i.e., for all  $j \in \{1, \dots, N\}$

$$|a_{j,j}| > \sum_{k \neq j} |a_{j,k}|. \quad (195)$$

Apart from this, the convergence condition for the Gauss-Seidel method holds if  $A$  is a positive definite matrix. Then, these iterative methods achieve the following convergence guarantee.

**Corollary 6** (The number of iterations in iterative methods). *For any target error  $\epsilon > 0$ , if the iterative matrix  $C$  in (182) satisfies*

$$\|C\| < 1, \quad (196)$$

*then after  $T$  iterations with*

$$T = O\left(\log \frac{1}{\epsilon}\right). \quad (197)$$

*the iterative methods in (181) output  $x^{(T)}$  achieving*

$$\left\|x^{(T)} - x^*\right\| \leq \epsilon, \quad (198)$$

*where  $x^*$  is the solution.*

*Proof.* Theorem 5 shows that

$$\left\|x^{(T)} - x^*\right\| = \left\|C^T(x^{(0)} - x^*)\right\| \leq \|C\|^T \left\|x^{(0)} - x^*\right\|. \quad (199)$$

We set

$$T = \log_{\|C\|} \left[ \frac{\epsilon}{\left\|x^{(0)} - x^*\right\|} \right] = \frac{\ln \left[ \frac{\left\|x^{(0)} - x^*\right\|}{\epsilon} \right]}{\ln \left[ \frac{1}{\|C\|} \right]}. \quad (200)$$

Then, we have

$$\left\|x^{(T)} - x^*\right\| \leq \|C\|^T \left\|x^{(0)} - x^*\right\| \leq \epsilon. \quad (201)$$

Since  $\left\|x^{(0)} - x^*\right\|$  and  $\|C\|$  are constants, we have

$$T = O\left(\log \frac{1}{\epsilon}\right). \quad (202)$$

□

## IV. EIGENVALUE PROBLEMS

### A. Eigenvalues

Given an  $N \times N$  matrix  $A$ , an eigenvalue is a (real or complex) number  $\lambda$  such that the equation, for a length- $N$  vector  $x$ ,

$$Ax = \lambda x \quad (203)$$

has a nontrivial solution  $x \neq 0$ , which is called an eigenvector corresponding to the eigenvalue  $\lambda$ . The set of all eigenvalues of  $A$  is called the spectrum of  $A$ .

One can rewrite (203) as

$$(A - \lambda I)x = 0, \quad (204)$$

where  $I$  is the identity matrix. This system of linear equations has a nontrivial solution if and only if the characteristic determinant  $\det(A - \lambda I)$  is 0. The eigenvalues of  $A$  are the solutions  $\lambda$  of the characteristic equation

$$\det(A - \lambda I) = 0. \quad (205)$$

The left-hand side is a degree- $N$  polynomial in terms of  $\lambda$ , which is called a characteristic polynomial.

The characteristic polynomial has  $N$  roots since it is a degree- $N$  polynomial. We write the eigenvalues of  $A$  as

$$\lambda_1, \dots, \lambda_N, \quad (206)$$

where some (or all) of them are equal. Then, the characteristic function can be represented as

$$f(\lambda) := \det(A - \lambda I) = (-1)^N \prod_{n=1}^N (\lambda - \lambda_n). \quad (207)$$

If we take equal factors together and let  $\lambda_1, \dots, \lambda_R$  ( $R \leq N$ ) denote the distinct eigenvalues of  $A$ , we also have

$$f(\lambda) = (-1)^N \prod_{r=1}^R (\lambda - \lambda_r)^{m_r}, \quad (208)$$

where  $m_r$  is called the algebraic multiplicity of  $\lambda_r$ . The maximum number of linearly independent eigenvectors corresponding to  $\lambda_r$  is called the geometric multiplicity of  $\lambda_r$ , which is smaller than or equal to  $m_r$ .

Knowing eigenvalues is useful for analyzing numerical problems involving matrices. The trace of  $A$ , the sum of main diagonals, is given by

$$\text{Tr}[A] := \sum_{n=1}^N a_{n,n} = \sum_{n=1}^N \lambda_n. \quad (209)$$

The determinant of  $A$  is given by

$$\det(A) = \prod_{n=1}^N \lambda_n. \quad (210)$$

For an  $N \times N$  complex matrix  $A$ , a subspace  $S \subset \mathbb{C}^N$  (or a subspace of  $\mathbb{R}^N$  for an  $N \times N$  real matrix  $A$ ) is called an invariant subspace of  $A$  if for every vector  $v \in S$ , we have

$$Av \in S. \quad (211)$$

For an eigenvalue  $\lambda$  of  $A$ , its eigenspace is the subspace spanned by eigenvectors corresponding to the eigenvalue  $\lambda$ , which is an invariant subspace.

## B. Properties of eigenvalues

It is generally challenging to obtain eigenvalues by directly solving the characteristic equation (205) because these values are the roots of degree- $N$  polynomials. The following theorems provide bounds for eigenvalues.

An important theorem is the Gerschgorin circle theorem, which provides a region that includes the eigenvalues of a given square matrix. Given any  $N \times N$  complex matrix  $A$ , for each  $n \in \{1, \dots, N\}$ , a Gerschgorin disc  $D_n$  of the matrix  $A$  is defined as a closed circular region in the complex plane

$$D_n := \{z \in \mathbb{C} : |z - a_{n,n}| \leq R_n\}, \quad (212)$$

where the radius  $R_n$  of  $D_n$  is given by the sum of the absolute values of the non-diagonal entries in the  $n$ th row

$$R_n := \sum_{n' \neq n} |a_{n,n'}|. \quad (213)$$

**Theorem 7** (Gerschgorin Circle Theorem). *Given any  $N \times N$  matrix  $A \in \mathbb{C}^{N \times N}$ , for any eigenvalue  $\lambda$  of  $A$ , there exists  $n \in \{1, \dots, N\}$  such that the eigenvalue  $\lambda$  is included in a Gerschgorin disc*

$$\lambda \in D_n, \quad (214)$$

where  $D_n$  is defined in (212).

*Proof.* Let

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix} \quad (215)$$

be an eigenvector corresponding to the eigenvalue  $\lambda$ , i.e.,  $Ax = \lambda x$  and  $x \neq 0$ . Let  $n$  be the index such that  $|x_n|$  is the largest among  $|x_1|, \dots, |x_N|$ . The characteristic equation yields

$$(A - \lambda I)x = 0, \quad (216)$$

which is a system of linear equations. The  $n$ th linear equation is

$$a_{n,1}x_1 + \dots + a_{n,n-1}x_{n-1} + (a_{n,n} - \lambda)x_n + a_{n,n+1}x_{n+1} + \dots + a_{n,N}x_N = 0. \quad (217)$$

Since  $x \neq 0$ , we have  $x_n \neq 0$ ; then the division by  $x_n$  and reshuffling terms gives

$$a_{n,n} - \lambda = -a_{n,1} \frac{x_1}{x_n} - \dots - a_{n,n-1} \frac{x_{n-1}}{x_n} - a_{n,n+1} \frac{x_{n+1}}{x_n} - \dots - a_{n,N} \frac{x_N}{x_n}. \quad (218)$$

Therefore, we have

$$|a_{n,n} - \lambda| = \left| -a_{n,1} \frac{x_1}{x_n} - \dots - a_{n,n-1} \frac{x_{n-1}}{x_n} - a_{n,n+1} \frac{x_{n+1}}{x_n} - \dots - a_{n,N} \frac{x_N}{x_n} \right| \quad (219)$$

$$\leq \left| a_{n,1} \frac{x_1}{x_n} \right| + \dots + \left| a_{n,n-1} \frac{x_{n-1}}{x_n} \right| + \left| a_{n,n+1} \frac{x_{n+1}}{x_n} \right| + \dots + \left| a_{n,N} \frac{x_N}{x_n} \right| \quad (220)$$

$$\leq \sum_{n' \neq n} |a_{n,n'}|, \quad (221)$$

where the last line follows from  $|x_1|/|x_n| \leq 1, \dots, |x_N|/|x_n| \leq 1$ .  $\square$

The following theorem provides a stronger guarantee on the eigenvalues.

**Theorem 8** (The number of eigenvalues in Gerschgorin discs). *For any  $N \in \{2, 3, \dots\}$ ,  $p \in \{1, \dots, N-1\}$ , and any  $N \times N$  matrix  $A \in \mathbb{C}^{N \times N}$ , suppose that  $p$  Gerschgorin discs of the matrix  $A$  form a set  $S$  that is disjoint from the  $N-p$  other Gerschgorin discs of  $A$ . Then,  $S$  contains precisely  $p$  eigenvalues of  $A$ , where each eigenvalue is counted with its algebraic multiplicity.*

*Proof.* Let  $D$  be a diagonal matrix whose diagonal elements are the diagonal elements  $a_{n,n}$  of  $A$  for all  $n \in \{1, \dots, N\}$ . For  $\epsilon \in [0, 1]$ , we consider a matrix  $B(\epsilon)$  parameterized by  $\epsilon$

$$B(\epsilon) := (1 - \epsilon)D + \epsilon A, \quad (222)$$

where  $B(0) = D$ , and  $B(1) = A$ . Each of the eigenvalues of  $B(0)$  is the center of one of the Gerschgorin discs of  $A$ ; thus, exactly  $p$  of the eigenvalues of  $B(\epsilon)$  lie in the union of the Gerschgorin discs in  $S$ . For any  $\epsilon$ , the eigenvalues of  $B(\epsilon)$  are the roots of its characteristic polynomial, which is a polynomial whose coefficients are continuous functions of  $\epsilon$ ; hence, the roots of this polynomial are also continuous functions of  $\epsilon$ . As  $\epsilon$  increases from 0 to 1, the eigenvalues of  $B(\epsilon)$  move along continuous paths in the complex plane, and at the same time, the radii of the Gerschgorin discs increase from 0 to the radii of the Gerschgorin discs of  $A$ . Since  $p$  of the eigenvalues lie in the union of the Gerschgorin discs in  $S$  when  $\epsilon = 0$ , and these discs are disjoint from all the  $N-p$  others, these  $p$  eigenvalues must still lie in the union of the Gerschgorin discs in  $S$  when  $\epsilon = 1$ .  $\square$

### C. Power methods

Given an  $N \times N$  diagonalizable matrix  $A$ , the power method provides an approximation of an eigenvalue and the corresponding eigenvector of  $A$ . For a matrix  $A$  and a vector  $v$ , the Rayleigh quotient of  $A$  and  $v$  is defined as

$$\frac{v^\dagger A v}{v^\dagger v}. \quad (223)$$

If  $v$  is an eigenvector corresponding to an eigenvalue  $\lambda$  of  $A$ , then the Rayleigh quotient of  $v$  becomes the eigenvalue  $\lambda$ . Based on this, the power method approximates the eigenvector corresponding to the eigenvalue with the largest absolute value. The power method proceeds as follows:

1. choose an initial vector  $x^{(0)}$ ;
2. for  $t \in \{1, \dots, T\}$ , iterate the following update:

$$x^{(t)} = Ax^{(t-1)}; \quad (224)$$

3. after the  $T$  iterations, return the Rayleigh quotient of  $x^{(T)}$

$$\lambda = \frac{x^{(T)\dagger}Ax^{(T)}}{x^{(T)\dagger}x^{(T)}}. \quad (225)$$

The power method guarantees the following.

**Theorem 9** (Power method). *For any  $N \times N$  diagonalizable matrix  $A$  with eigenvalues  $\lambda_1, \dots, \lambda_N$  satisfying*

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_N|, \quad (226)$$

*if the initial vector  $x^{(0)}$  in the power method has a nonzero overlap with a normalized eigenvector  $v_1$  ( $\|v_1\| = 1$ ) corresponding to the dominant eigenvalue  $\lambda_1$ , i.e.,*

$$x^{(0)\dagger}v_1 \neq 0, \quad (227)$$

*then a vector  $x \propto x^{(T)}$  obtained from the power method with  $T$  iterations satisfies*

$$\|x - v_1\| = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^T\right), \quad (228)$$

where  $\|\dots\|$  is the Euclidean norm.

*Proof.* Since  $A$  is diagonalizable, normalized eigenvectors  $v_1, \dots, v_N$  corresponding to the eigenvalues  $\lambda_1, \dots, \lambda_N$  form a basis, where their Euclidean norm satisfies  $\|v_1\| = \dots = \|v_N\| = 1$ . Using this basis, the initial vector  $x^{(0)}$  is expanded as

$$x^{(0)} = c_1v_1 + c_2v_2 + \dots + c_Nv_N, \quad (229)$$

where  $c_1 \neq 0$  due to (227). Applying the power method for  $T$  iterations, we obtain

$$x^{(1)} = Ax^{(0)} = c_1\lambda_1v_1 + c_2\lambda_2v_2 + \dots + c_N\lambda_Nv_N, \quad (230)$$

$$x^{(2)} = A^2x^{(0)} = c_1\lambda_1^2v_1 + c_2\lambda_2^2v_2 + \dots + c_N\lambda_N^2v_N, \quad (231)$$

$$\vdots \quad (232)$$

$$x^{(T)} = A^Tx^{(0)} = c_1\lambda_1^Tv_1 + c_2\lambda_2^Tv_2 + \dots + c_N\lambda_N^Tv_N. \quad (233)$$

Hence, we have

$$\frac{1}{c_1\lambda_1^T}x^{(T)} = v_1 + \frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^Tv_2 + \dots + \frac{c_N}{c_1}\left(\frac{\lambda_N}{\lambda_1}\right)^Tv_N. \quad (234)$$

We have

$$\left\|\frac{1}{c_1\lambda_1^T}x^{(T)} - v_1\right\| = \left\|\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^Tv_2 + \dots + \frac{c_N}{c_1}\left(\frac{\lambda_N}{\lambda_1}\right)^Tv_N\right\| \quad (235)$$

$$= \frac{1}{c_1}\sqrt{|c_2|^2\left|\frac{\lambda_2}{\lambda_1}\right|^{2T} + \dots + |c_N|^2\left|\frac{\lambda_N}{\lambda_1}\right|^{2T}} \quad (236)$$

$$\leq \frac{1}{c_1}\left|\frac{\lambda_2}{\lambda_1}\right|^T\sqrt{|c_2|^2 + \dots + |c_N|^2} \quad (237)$$

$$\leq \frac{1}{c_1}\left|\frac{\lambda_2}{\lambda_1}\right|^T\sqrt{|c_1|^2 + |c_2|^2 + \dots + |c_N|^2} \quad (238)$$

$$= \frac{\|x^{(0)}\|}{c_1}\left|\frac{\lambda_2}{\lambda_1}\right|^T. \quad (239)$$

□

Instead of (224), one may perform an update by solving

$$Ax^{(t)} = x^{(t-1)} \quad (240)$$

using the Gaussian elimination. Since this is equivalent to the update by  $x^{(t)} = A^{-1}x^{(t-1)}$ , the algorithm returns an approximation of the eigenvector corresponding to the eigenvalue of  $A^{-1}$  with the largest absolute value, i.e., the eigenvalue of  $A$  with the smallest absolute value. Furthermore, for any  $N \times N$  matrix  $A$  with eigenvalues  $\lambda_1, \dots, \lambda_N$ , the matrix  $A - \lambda I$  has eigenvalues  $\lambda_1 - \lambda, \dots, \lambda_N - \lambda$ , while the eigenvectors remain unchanged. Using this, for any  $\lambda$  such that  $A - \lambda I$  is invertible, instead of (224), one may perform the update by solving

$$(A - \lambda I)x^{(t)} = x^{(t-1)} \quad (241)$$

using the Gaussian elimination. This provides the eigenvector corresponding to the eigenvalue of  $A$  that is the closest to  $\lambda$ . This method is called the inverse power method.

## V. METHODS FOR COMPUTING ALL EIGENVALUES

The QR algorithm is an algorithm to compute all the eigenvalues of a real symmetric matrix  $A \in \mathbb{R}^{N \times N}$ , i.e.,

$$A^\top = A, \quad (242)$$

where  $A^\top$  is the transpose of  $A$ . An  $N \times N$  real matrix  $Q$  is called an orthogonal matrix if it holds that

$$Q^{-1} = Q^\top. \quad (243)$$

Given an  $N \times N$  real matrix  $A$ , a QR decomposition of  $A$  is

$$A = QR, \quad (244)$$

where  $Q$  is an orthogonal matrix, and  $R$  is an upper triangular matrix. If  $A$  is an invertible real matrix and the main diagonals of  $R$  are restricted to be positive, then the QR decomposition of  $A$  is unique. In this section, the QR algorithm to compute all the eigenvalues of  $A$  is presented based on the QR decomposition.

### A. Householder's tridiagonalization method

A tridiagonal matrix is a matrix having all its nonzero elements on the main diagonals and the elements immediately adjacent to the main diagonals:

$$\begin{pmatrix} * & * & 0 & 0 & 0 \\ * & * & * & 0 & 0 \\ 0 & * & * & * & 0 \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{pmatrix}. \quad (245)$$

For any  $N \times N$  matrix  $A$ , an  $N \times N$  matrix  $B$  is called similar to  $A$  if there exists an  $N \times N$  invertible matrix  $T$  such that

$$B = T^{-1}AT. \quad (246)$$

Similar matrices have the same eigenvalues; in particular, if  $x$  is an eigenvector of  $A$ , then  $y = T^{-1}x$  is an eigenvector of  $B$  corresponding to the same eigenvalue. We will transform  $A$  into similar matrices to identify its eigenvalues. For a real symmetric matrix  $A \in \mathbb{R}^{N \times N}$ , the Householder tridiagonalization method is a technique to obtain a tridiagonal matrix  $B$  that is similar to  $A$ , thereby simplifying the QR decomposition.

Given any  $N \times N$  real symmetric matrix  $A$ , the Householder tridiagonalization method proceeds as follows:

1. set  $A^{(0)} := A$ ;

2. for each step  $t \in \{1, 2, \dots, N-2\}$ , define

$$P_t := I - 2v_t v_t^\top, \quad (247)$$

where

$$v_t = \begin{pmatrix} v_{1,t} \\ v_{2,t} \\ \vdots \\ v_{N,t} \end{pmatrix} \quad (248)$$

is a normalized vector given, for  $A^{(t-1)} = (a_{i,j}^{(t-1)})_{i,j}$ , by

$$v_{1,t} = \dots = v_{t,t} = 0, \quad (249)$$

$$v_{t+1,t} = \sqrt{\frac{1}{2} \left( 1 + \frac{|a_{t+1,t}^{(t-1)}|}{S_t} \right)}, \quad (250)$$

$$v_{n,t} = \frac{a_{n,t} \operatorname{sgn}(a_{t+1,t}^{(t-1)})}{2v_{t+1,t} S_t}, \quad n \in \{t+1, \dots, N\}, \quad (251)$$

$\operatorname{sgn}(a_{t+1,t}^{(t-1)}) \in \{\pm 1\}$  is the sign of the element  $a_{t+1,t}^{(t-1)}$  of  $A^{(t-1)}$  computed in the previous step, and  $S_t$  is given by

$$S_t = \sqrt{a_{t+1,t}^{(t-1)2} + a_{t+2,t}^{(t-1)2} + \dots + a_{N,t}^{(t-1)2}}. \quad (252)$$

With this  $P_t$ , we compute

$$A^{(t)} = P_t A^{(t-1)} P_t; \quad (253)$$

3. finally, output

$$B = A^{(N-2)}, \quad (254)$$

which becomes a real symmetric tridiagonal matrix.

For example, consider

$$A^{(0)} = A = \begin{pmatrix} -1 & -1 & 2 & 2 \\ -1 & 2 & 1 & 1 \\ 2 & 1 & 2 & 1 \\ 2 & 1 & 1 & 2 \end{pmatrix}. \quad (255)$$

We have

$$S_1 = 3, \quad (256)$$

$$v_1 = \begin{pmatrix} 0 \\ \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \end{pmatrix} \quad (\text{the first element becomes } 0), \quad (257)$$

$$P_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \\ 0 & \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix} \quad (\text{the first row and column become } I), \quad (258)$$

$$A^{(1)} = \begin{pmatrix} -1 & 3 & 0 & 0 \\ 3 & 2 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix} \quad (\text{the first row and column becomes tridiagonal}), \quad (259)$$

and

$$S_2 = \sqrt{2}, \quad (260)$$

$$v_2 = \begin{pmatrix} 0 \\ 0 \\ \frac{\sqrt{2+\sqrt{2}}}{2} \\ \frac{\sqrt{2-\sqrt{2}}}{2} \end{pmatrix} \quad (\text{the first and second elements become } 0), \quad (261)$$

$$P_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad (\text{the first and second rows and columns become } I), \quad (262)$$

$$A^{(2)} = \begin{pmatrix} -1 & 3 & 0 & 0 \\ 3 & 2 & -\sqrt{2} & 0 \\ 0 & -\sqrt{2} & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{the first and second rows and columns become triorthogonal}). \quad (263)$$

**Theorem 10.** *For any  $N \times N$  real symmetric matrix  $A$ , the triorthogonal matrix  $B$  obtained from the Householder tridiagonalization method is similar to  $A$ ; that is,  $A$  and  $B$  have the same eigenvalues.*

*Proof.* By construction,  $P_t$  is a symmetric matrix:

$$P_t^\top = (I - 2v_t v_t^\top)^\top = I^\top - 2(v_t v_t^\top)^\top = I - 2v_t v_t^\top = P. \quad (264)$$

Since  $v_t$  is a unit vector,  $P_t$  is an orthogonal matrix:

$$P_t P_t^\top = P_t^2 = (I - 2v_t v_t^\top)^2 = I - 2(2v_t v_t^\top) + 4v_t v_t^\top v_t v_t^\top = I. \quad (265)$$

Due to (264) and (265), we have

$$P_t^{-1} = P_t^\top = P_t. \quad (266)$$

Therefore, we have

$$B = P_{N-2} A^{(N-3)} P_{N-2} \quad (267)$$

$$= P_{N-2} P_{N-3} A^{(N-4)} P_{N-3} P_{N-2} \quad (268)$$

$$= \dots \quad (269)$$

$$= P_{N-2} P_{N-3} \dots P_1 A P \dots P_{N-3} P_{N-2} \quad (270)$$

$$= P A P, \quad (271)$$

where  $P := P_{N-2} P_{N-3} \dots P_1$  is an invertible matrix.  $\square$

## B. QR decomposition and QR algorithm

Given a real symmetric matrix  $A \in \mathbb{R}^{N \times N}$ , the QR algorithm is an iterative algorithm using the QR decomposition to compute all the eigenvalues of  $A$ . The QR algorithm proceeds as follows:

1. compute a tridiagonal matrix  $B_0$  that is similar to  $A$ , by using the Householder tridiagonalization method;
2. for each  $t \in \{1, \dots, T\}$ , compute the QR decomposition of  $B_{t-1}$  (as shown below)

$$B_{t-1} = Q_{t-1} R_{t-1}, \quad (272)$$

and compute an updated matrix

$$B_t = R_{t-1} Q_{t-1}, \quad (273)$$

where one can ensure that  $B_t$  is a real symmetric tridiagonal matrix if  $B_{t-1}$  is a real symmetric tridiagonal matrix;

3. output the matrix  $B_T$ .

It is known that we have

$$B_T \approx D \text{ as } T \rightarrow \infty, \quad (274)$$

where  $D$  is a diagonal matrix with its main diagonals  $\lambda_1, \dots, \lambda_N$ . The proof of this fact can be found, e.g., in Ref. [5].

The algorithm to obtain the QR decomposition of an  $N \times N$  real symmetric tridiagonal matrix  $B$  is as follows. The tridiagonal matrix  $B = (b_{i,j})_{i,j}$  has at most  $N - 1$  nonzero elements below the main diagonal, which are  $b_{2,1}, b_{3,2}, \dots, b_{N,N-1}$ . We multiply  $B$  from the left by a matrix  $C_2$  to obtain  $C_2 B = (b_{i,j}^{(2)})_{i,j}$  with

$$b_{2,1}^{(2)} = 0. \quad (275)$$

We further multiply this by a matrix  $C_3$  to obtain  $C_3 C_2 B = (b_{i,j}^{(3)})_{i,j}$  with

$$b_{3,2}^{(3)} = 0. \quad (276)$$

By iterating this, we obtain

$$C_N C_{N-1} \cdots C_3 C_2 B = R, \quad (277)$$

where  $R$  is an upper triangular matrix. The  $N \times N$  matrices  $C_2, \dots, C_N$  to achieve this transformation is in the form of

$$C_2 = \begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & 0 & 0 & \cdots \\ \sin(\theta_2) & \cos(\theta_2) & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (278)$$

$$C_3 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & \cos(\theta_3) & -\sin(\theta_3) & 0 & 0 & \cdots \\ 0 & \sin(\theta_3) & \cos(\theta_3) & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (279)$$

$$C_4 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & \cos(\theta_4) & -\sin(\theta_4) & 0 & \cdots \\ 0 & 0 & \sin(\theta_4) & \cos(\theta_4) & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (280)$$

$$\vdots \quad (281)$$

and  $\theta_2, \theta_3, \dots$  are determined iteratively to satisfy (275), (276), ..., respectively. The matrices  $C_2, \dots, C_N$  are orthogonal matrices, and hence, by multiplying an orthogonal matrix

$$Q := C_2^\top C_3^\top \cdots C_{N-1}^\top C_N^\top \quad (282)$$

from the left in (277), we obtain the QR decomposition of  $B$

$$B = QR. \quad (283)$$

Due to the forms of the matrices  $C_2, \dots, C_N$  in (282), the updated matrix  $RQ$  in (273) of the QR algorithm is also tridiagonal. Since we have

$$B_t = R_{t-1} Q_{t-1} = Q_{t-1}^\top B_{t-1} Q_{t-1}, \quad (284)$$

and thus

$$B_t^\top = Q_{t-1}^\top B_{t-1}^\top Q_{t-1}, \quad (285)$$

$B_t$  is a real symmetric matrix if  $B_{t-1}$  is. Therefore, each iteration in the QR algorithm computes the QR decomposition of a real symmetric tridiagonal matrix  $B_{t-1}$  and updates it into a real symmetric tridiagonal matrix  $B_t$ .

## VI. SINGULAR VALUE DECOMPOSITION

Unlike the spectral decomposition of a square normal matrix, the singular value decomposition (SVD) is defined for any rectangular complex matrix. For any  $M \times N$  complex matrix  $A$ , the singular value decomposition of  $A$  is

$$A = U \Sigma V^\dagger, \quad (286)$$

where  $U$  is an  $M \times M$  unitary matrix  $UU^\dagger = U^\dagger U = I$ ,  $\Sigma$  is an  $M \times N$  rectangular diagonal matrix with nonnegative real numbers on the diagonal, and  $V^\dagger$  is the conjugate transpose of an  $N \times N$  unitary matrix  $V$ .

The diagonal elements

$$\sigma_n = \Sigma_{n,n} \geq 0 \quad (287)$$

of  $\Sigma$  are uniquely determined by  $A$  and are called the singular values of  $A$ . The columns  $u_1, \dots, u_M$  of  $U$  are called the left singular vectors of  $A$ , and the columns  $v_1, \dots, v_N$  of  $V$  are called the right singular vectors of  $A$ , which form the orthonormal bases. The number of nonzero singular values is equal to the rank of  $A$ . If the singular values are sorted in descending order, the singular value decomposition is written as

$$A = \sum_{n=1}^R \sigma_n u_n v_n^\dagger, \quad (288)$$

where  $R \leq \min\{M, N\}$  is the rank of  $M$ .

Since  $AA^\dagger$  and  $A^\dagger A$  are positive semidefinite matrices, they have spectral decomposition. With the singular value decomposition (288) of  $A$ , the spectral decomposition of  $AA^\dagger$  is given by

$$AA^\dagger = \sum_{n=1}^R \sigma_n^2 u_n u_n^\dagger, \quad (289)$$

and the spectral decomposition of  $A^\dagger A$  is given by

$$A^\dagger A = \sum_{n=1}^R \sigma_n^2 v_n v_n^\dagger. \quad (290)$$

Thus, by computing the eigenvalues and the eigenvectors of  $AA^\dagger$  and  $A^\dagger A$ , one can obtain the singular value decomposition of  $A$ .

The applications of the singular value decomposition include the following.

1. Low-rank approximation: For  $A$  in (288) with  $\sigma_1 \geq \sigma_2 \geq \dots$ ,  $A_k = \sum_{n=1}^k \sigma_n u_n v_n^\dagger$  is a rank- $k$  matrix minimizing  $\|A - A_k\|_F$ , where  $\|A\|_F = \text{Tr}[A^\dagger A] = \sum_{i,j} |a_{i,j}|^2 = \sum_n \sigma_n^2$  is the Frobenius norm.
2. The spectral norm of  $A$  is given by  $\|A\| = \max\{\sigma_n\}$ .
3. The condition number of  $A$  is given by  $\kappa = \max\{\sigma_n\} / \min\{\sigma_n\}$ .

## VII. SUMMARY

1. A system of linear equations. Determinant, minor, cofactor, and inverse matrices.
2. Gaussian elimination, and Gauss-Jordan elimination.
3. LU decomposition.
4. Vector space, vector norm, and distance.
5. Normal matrices, Hermitian matrices, positive semidefinite matrices, unitary matrices, and orthogonal matrices.
6. Spectral decomposition, eigenvalue, and eigenvectors.
7. Matrix norm. Spectral norm.

8. Condition number.
9. Conjugate gradient method.
10. Gauss-Seidel method and Jacobi method.
11. Gerschgorin Circle theorem.
12. Power method, and inverse power method.
13. Householder's tridiagonalization.
14. QR decomposition, and QR algorithm.
15. Singular value decomposition. Singular values, left singular vectors, and right singular vectors.

## VIII. APPLICATIONS

1. Harrow-Hassidim-Lloyd algorithm [6, 7].
2. Synthesis of gadgets for fault-tolerant quantum computation [8].
3. Efficient decoding in quantum error correction [9, 10].
4. Analysis of machine learning [11, 12].

See also <https://www.hayatayamasaki.com/research/%E7%A0%94%E7%A9%B6%E7%B4%B9%E4%BB%8B-%E6%97%A5%E6%9C%AC%E8%AA%9E> for more details.

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